# House Price Prediction System

**Project Id: PY-PT48**


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# 1. Problem Description

The goal of this project is to develop a robust machine learning model to predict residential property prices based on physical attributes and environmental factors. By leveraging historical data, the system provides an automated "Official Valuation Report" to assist buyers and sellers in estimating fair market value.

## 1.1 Dataset Description

The dataset used in this project contains information required for prediction/recommendation tasks.

**Dataset Features**

The dataset includes:

- User-related information
- Item/Movie/Product information
- Ratings or target values
- Historical behavioral patterns

**Dataset Characteristics**

- Dataset Format: CSV
- Number of Records: [1460]
- Number of Features: [30]
- Target Variable: [SALE PRICE]

**Data Cleaning Performed**

The following preprocessing steps were applied:

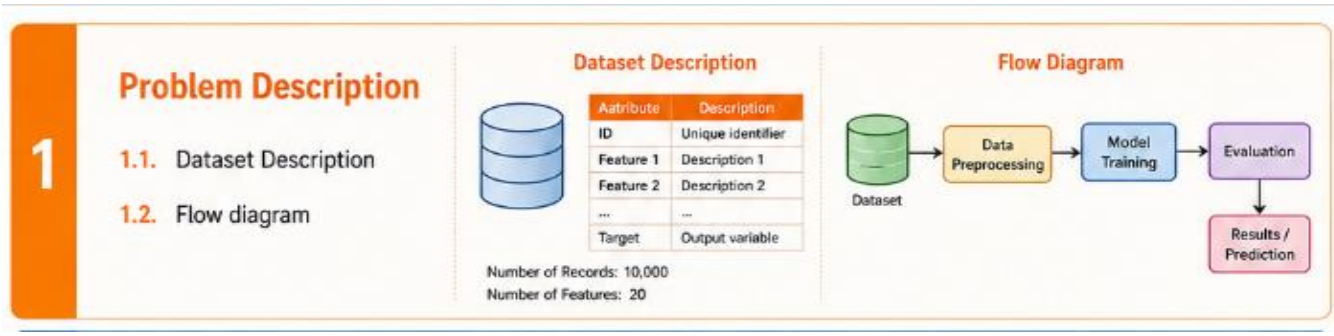
- Handling missing values
- Removing duplicate records
- Encoding categorical values
- Feature scaling/normalization (if applied)
- Log transformation of target variable (if regression project)

**Data Splitting**

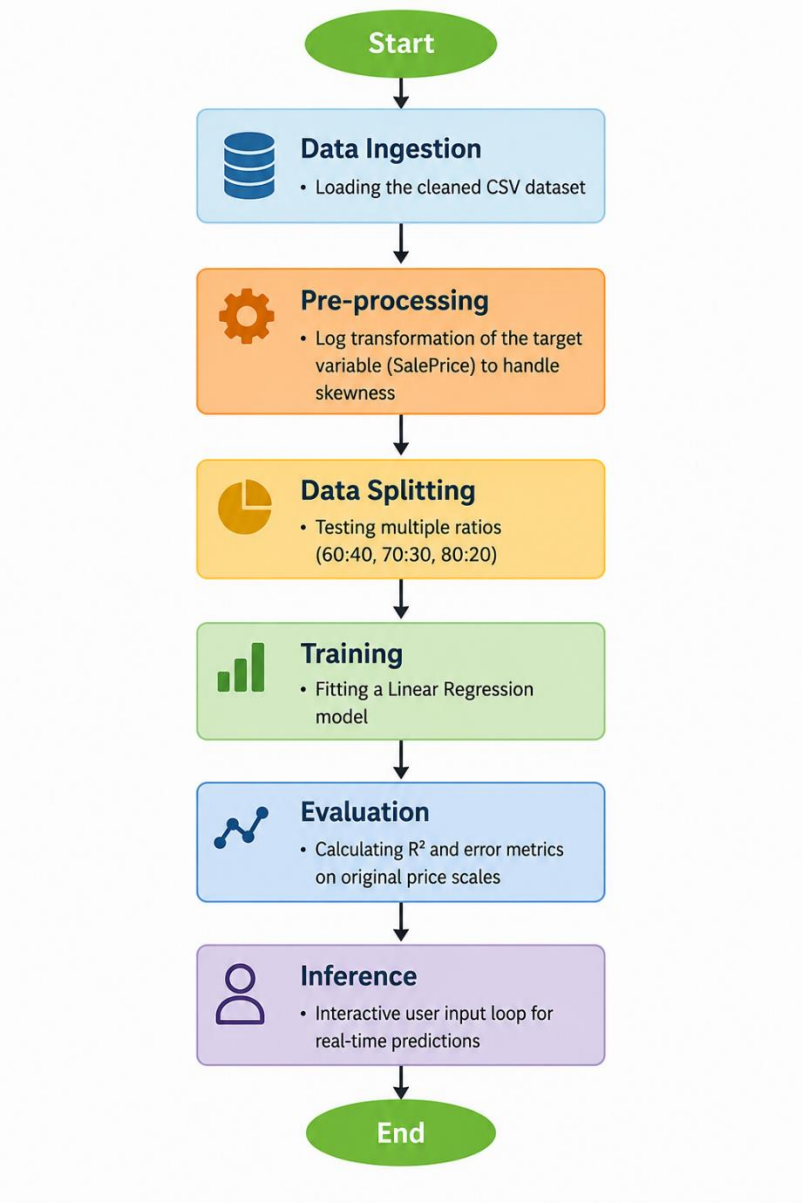
The dataset was divided into:

- Training Data: 80%
- Testing Data: 20%

Other train-test ratios such as 60:40 and 70:30 were also tested to compare model performance.



## 1.2 Flow Diagram




- **Data Ingestion:** Loading the cleaned CSV dataset.
- **Pre-processing:** Log transformation of the target variable (SalePrice) to handle skewness.
- **Data Splitting:** Testing multiple ratios (60:40, 70:30, 80:20).
- **Training:** Fitting a Linear Regression model.
- **Evaluation:** Calculating R2 score and error metrics on original price scales.
- **Inference:** Interactive user input loop for real-time predictions.

# 2. Modules for Project Implementation

## 2.1 Modules Used

- **Pandas:** For data manipulation and loading CSV files.
- **NumPy:** For mathematical operations, specifically log transformations (np.log1p) and exponentiation (np.expm1).
- **Scikit-Learn:**
- train_test_split: For dividing data.
- LinearRegression: The core algorithm.
- metrics: To calculate R2 score, MAE, and MSE.
- **Matplotlib/Seaborn:** For generating comparison bar charts and scatter plots.

# Description of Important Modules

**NumPy**
Used for mathematical operations and array handling.
**Pandas**
Used for loading CSV files, cleaning data, and manipulating datasets.
**Matplotlib and Seaborn**
Used for creating graphs and visualizations.
**Scikit-learn**
Provides Machine Learning models, preprocessing tools, and evaluation metrics.

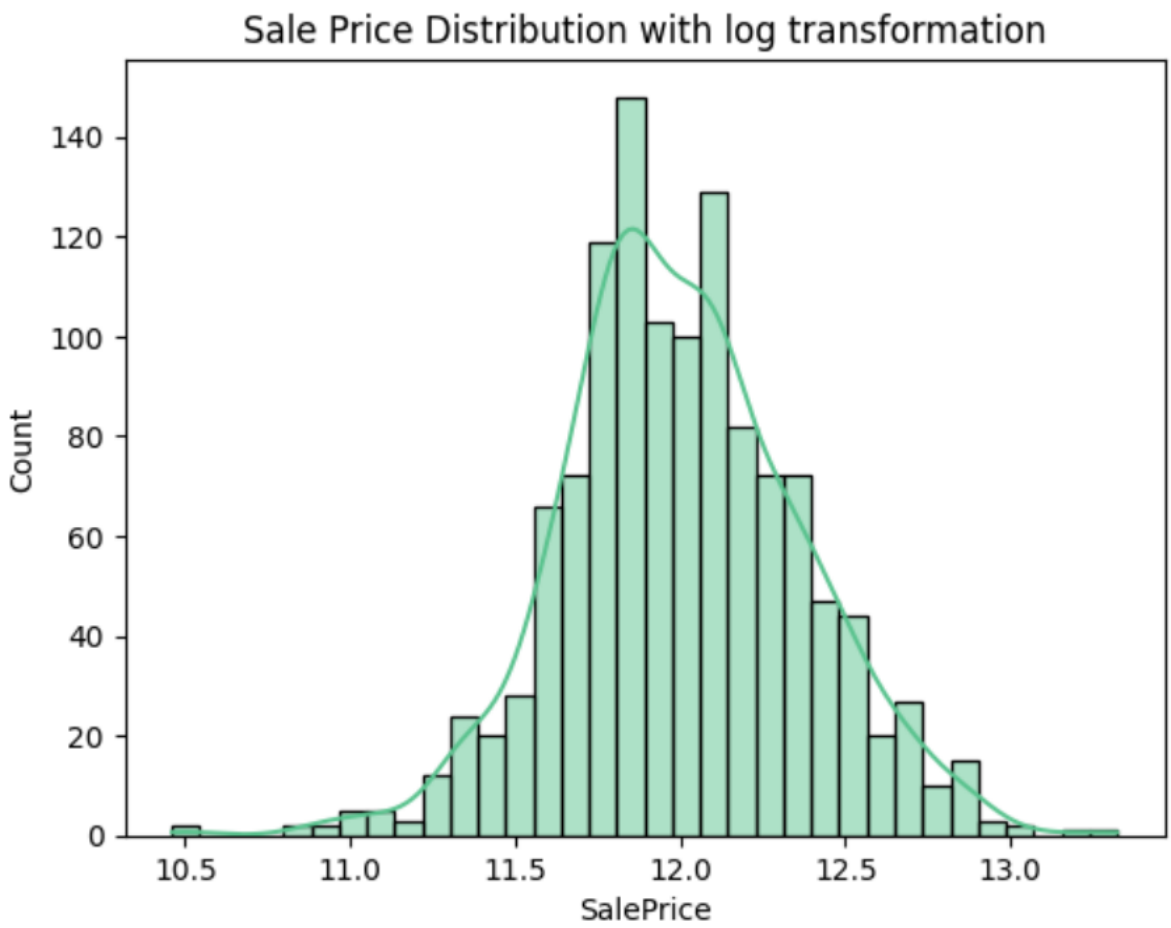
## 2.2 Platform Used:

- Environment: JupyterLab / Jupyter Notebook.
- Language: Python 3 (ipykernel).

## 3. Machine Learning Model/ Searching Technique Used

- The project utilizes Multiple Linear Regression. To improve model performance and stabilize variance, a Log-Linear approach was implemented:

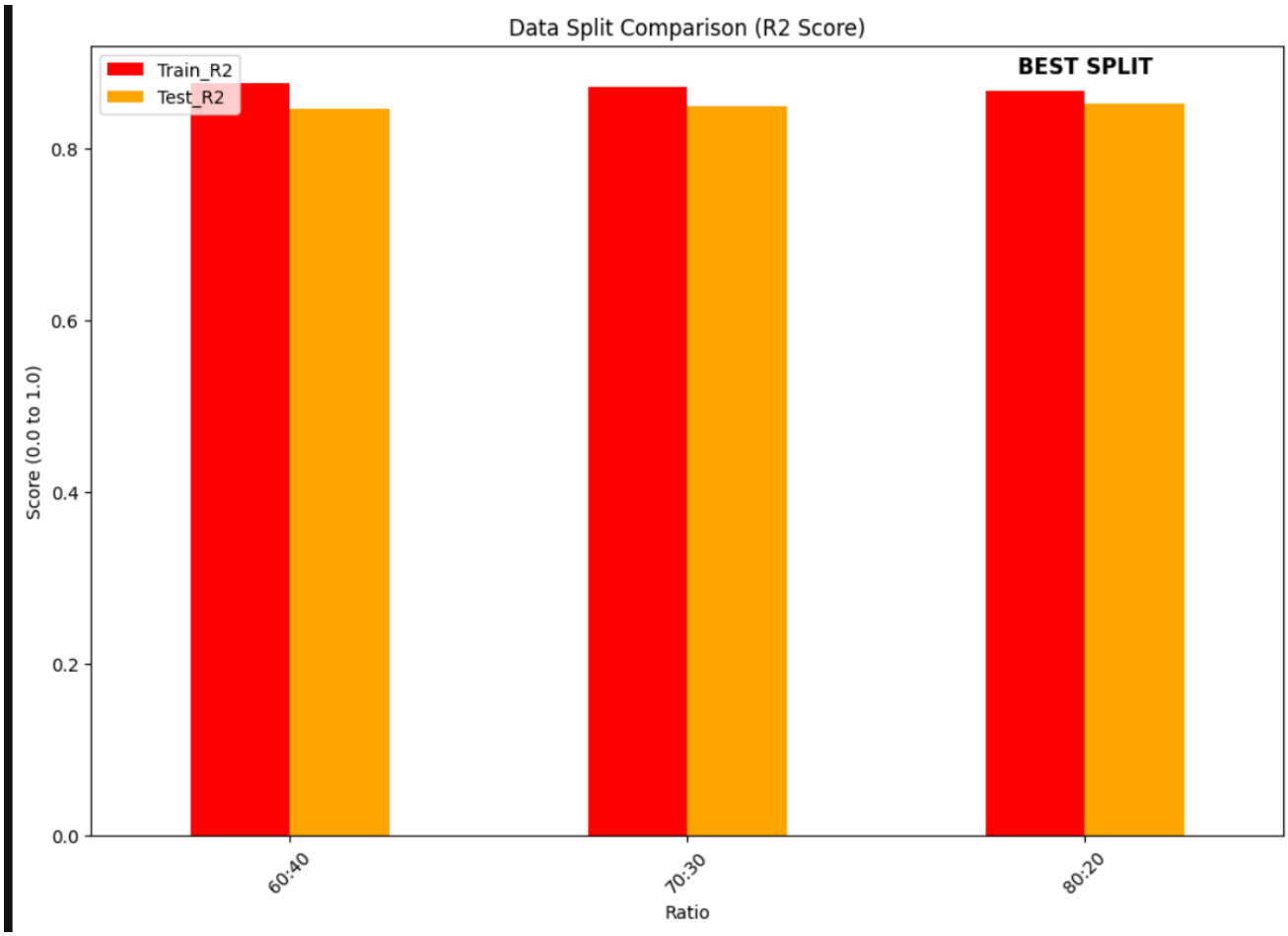
- This ensures that the model predicts the natural log of the price, which is then converted back to the original dollar amount for the final report.



## 4. Evaluation metrics used

The model was evaluated using four primary metrics:

- R-Squared score : Measures the proportion of variance explained by the model (achieved ~0.85 on test data).
- Mean Absolute Error (MAE): Represents the average dollar-amount error ($18,541 in the best split).
- Mean Squared Error (MSE/RMSE): Penalizes larger errors to assess model reliability.



# 5. Results and Visualizations

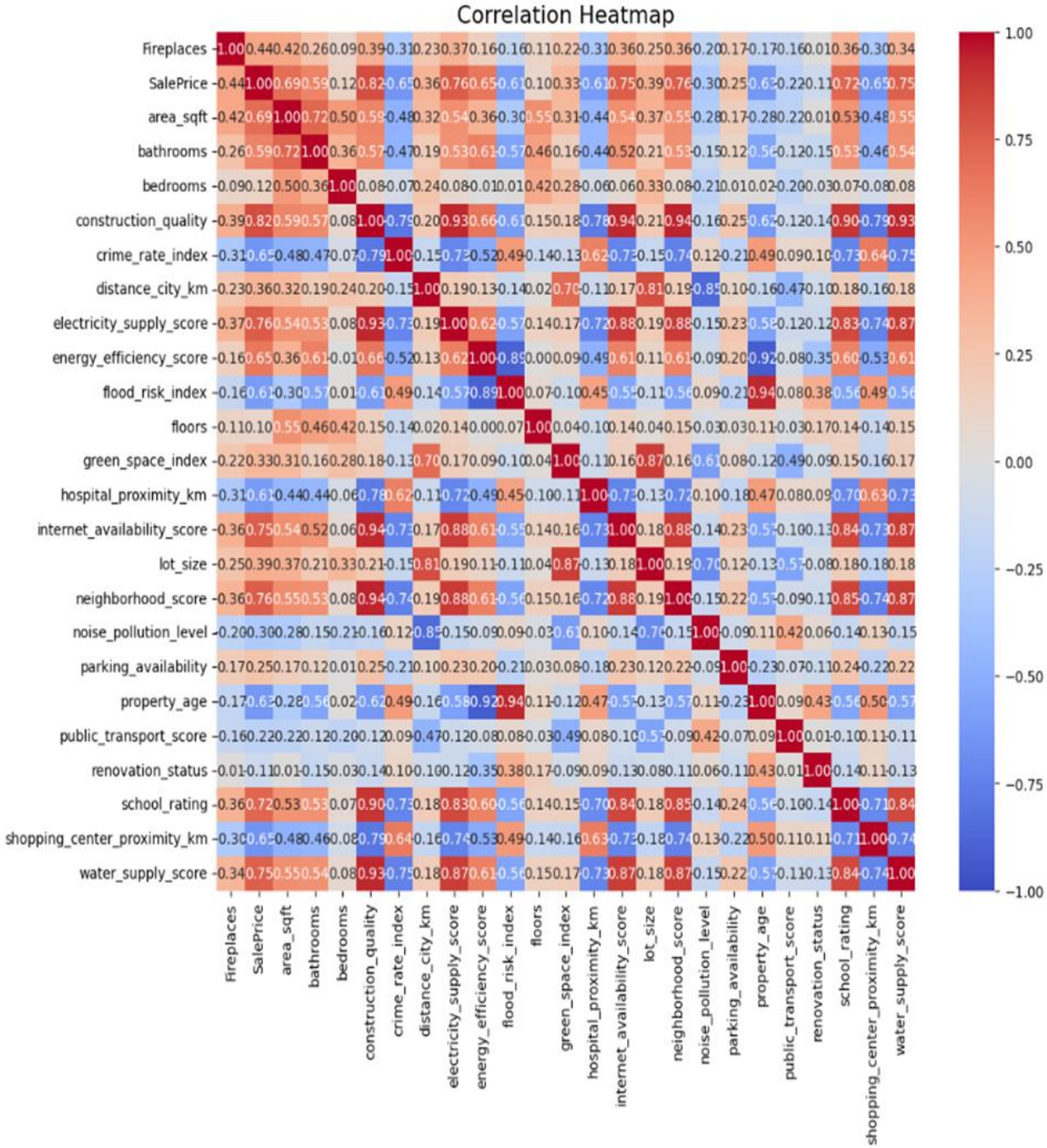



Fig.2 Heatmap showing correlation.

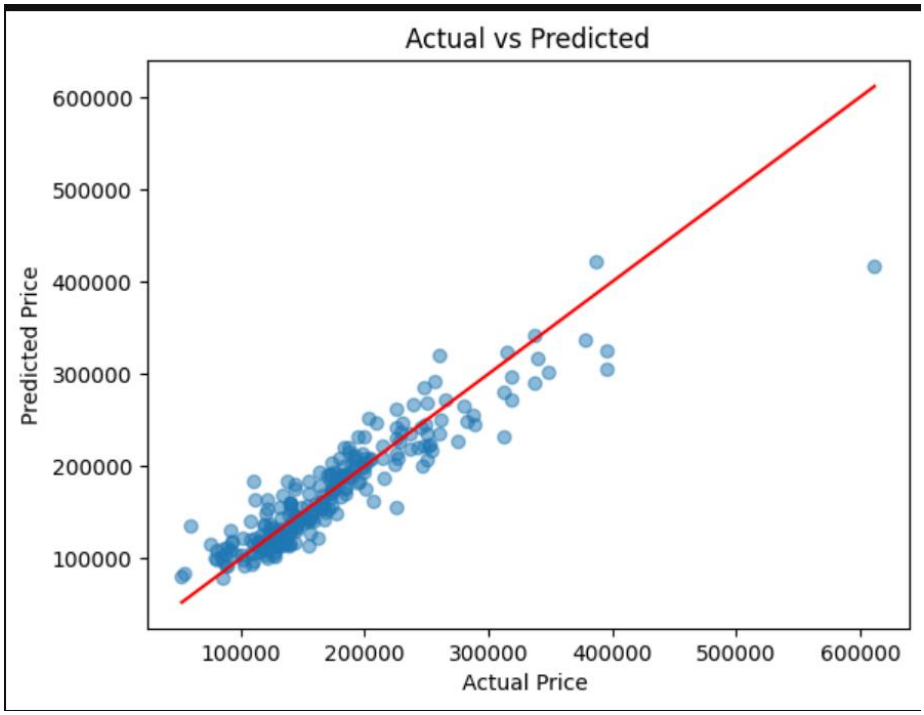



Fig.LINEAR REGRESSION

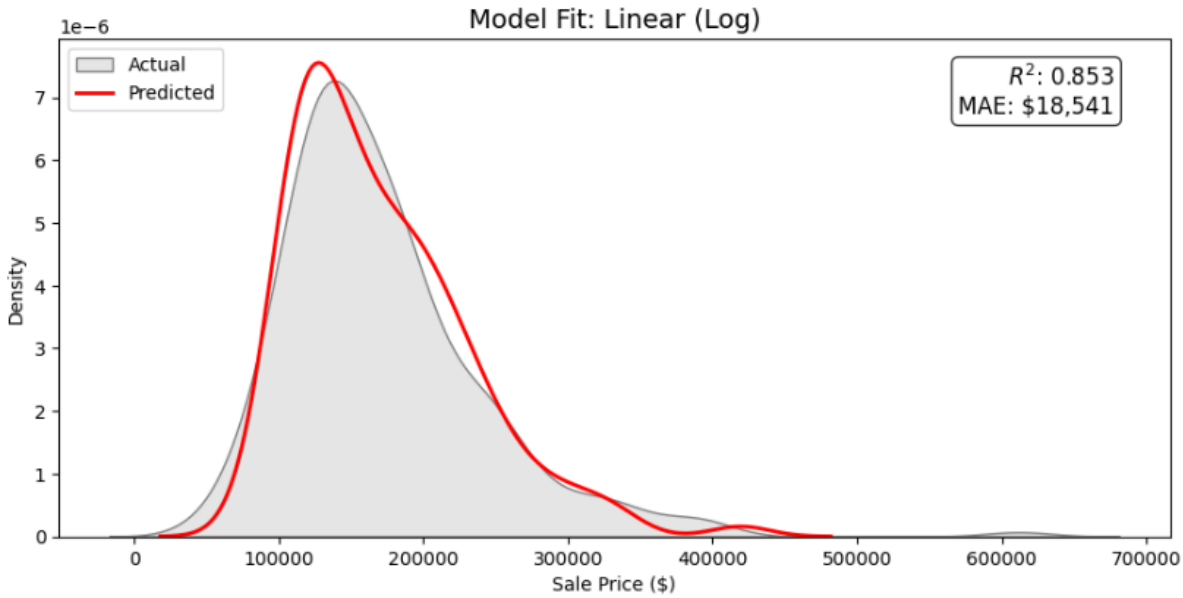



Fig.PREDICTED VS ACTUAL VALUES

```
🏠 House Valuation System (Validation Enabled)
-----------------------------------------------------
➤ Enter area (500-5000):  3500
➤ Number of bedrooms (1-6):  3
➤ Number of bathrooms (1-5):  3
➤ Number of floors (1-3):  2
➤ Age in years (0 for new):  5
➤ Renovation (0: None, 10: Full):  7
➤ Enter Lot Size (Must be at least 3500.0 sqft):  5000
➤ Neighborhood (1-10):  8
➤ School quality (1-10):  7
➤ Parking? (1: Yes, 0: No):  1
➤ Build quality (1-10):  6
➤ Water reliability (1-10):  9
➤ Parks/Nature (1-10):  8
➤ Fireplaces (0, 1, or 2):  2
➤ Is it a Duplex? (1: Yes, 0: No):  0

==============================================
         OFFICIAL VALUATION REPORT
==============================================
Property Feature          | Input Value
----------------------------------------------
Area Sqft                 | 3500
Bedrooms                  | 3
Bathrooms                 | 3
Floors                    | 2
Property Age              | 5
Renovation Status         | 7
Lot Size                  | 5000
Neighborhood Score        | 8
School Rating             | 7
Parking Availability      | 1
Construction Quality      | 6
Water Supply Score        | 9
Green Space Index         | 8
Fireplaces                | 2
Property Type Duplex      | 0
----------------------------------------------
ESTIMATED MARKET VALUE: $495,113.23
LIKELY PRICE RANGE:    $476,571.86 - $513,654.60
----------------------------------------------
```

# 6. Conclusion and Future Scope:

Conclusion

The Linear Regression model effectively predicts house prices with an $R^2$ of 0.853, indicating high reliability. The inclusion of quality-of-life metrics (schools, green space) significantly contributes to valuation accuracy.

Future Scope

- Algorithm Expansion: Implementing Random Forest or XGBoost to capture non-linear relationships.
- Deployment: Turning the notebook into a web application using Streamlit or Flask.
- Geospatial Data: Integrating GPS coordinates to account for hyper-local market trends.

# 7. References:

## Books:

1. *Let Us Python* – Yashavant Kanetkar

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## Online Resources:

- GeeksforGeeks – https://www.geeksforgeeks.org/
- TutorialsPoint – https://www.tutorialspoint.com/python/index.htm
- Stack Overflow – https://stackoverflow.com
- Kaggle – https://www.kaggle.com/
- ChatGPT – https://www.chatgpt.com
- Scikit-Learn Documentation: Linear Models
- Pandas API Reference: DataFrames
- Project Source: Phase_3.ipynb internal valuation logic.